

PROJECT OS CLI V3 — Reference & Next-Phase Planning

Handoff for ChatGPT web — 50/50 features PASS, ready for the next 50 development phases.

1. Identity & Location

- **Repo**: C:\Users\eiden\Desktop\dev\projet-os
- **CLI source**: cli-cpp/ (C++17, CMake, header-only helpers)
- **Binary**: cli-cpp/cmake-build/project-os-cli.exe
- **Build**: CMakePresets (Ninja): debug, release, test, workflow ci; installable (cmake --install ... --prefix)
- **Node bridge**: bin/project-os-bridge.mjs (protocol v2)
- **LocalAI**: http://127.0.0.1:8080/v1 (loopback, 15 models)
- **Toolchain**: MinGW C:\msys64\mingw64\bin, CMake 4.3.3, g++ 16.1.0
- **GPU**: NVIDIA RTX 5060 Laptop, 8 GB, driver 616.56

2. Architecture (hub-spoke)

- CLI (C++) parses/UX/terminal → runner (CreateProcessW, no shell) → Node bridge → SlashCommandRegistry → Project OS TS domain.
- The CLI **delegates** business logic to the bridge; it never re-implements Project OS.
- Layered: UI/extension → application/domain → services → providers/adapters (LocalAI/fs/bridge/GPU).

3. Baseline (real, measured)

- C++ build PASS · CTest **1/1 PASS** · typecheck **0 errors** · CLI soak **13/13** scenarios PASS.
- Feature matrix: **50/50 PASS**.

4. The 50 features (all implemented + PASS)

F01 version · F02 capabilities · F03 exit codes · F04 multi-format (human/json/ndjson/tsv) ·
 F05 terminal detection · F06 unicode · F07 process runner (CreateProcessW) · F08 timeout ·
 F09 ctrl+c · F10 protocol schema · F11 status · F12 project query · F13 project use · F14 project
 inspect ·
 F15 project watch · F16 drift · F17 timeline · F18 snapshot · F19 diff · F20 explain/dry-run ·
 F21 goal proof · F22 todo board · F23 artifact list · F24 artifact show · F25 artifact search ·
 F26 artifact verify · F27 addon verify · F28 config · F29 doctor · F30 diagnostics bundle ·
 F31 preflight · F32 health · F33 models · F34 model show · F35 route · F36 model smoke ·
 F37 model benchmark · F38 gpu status · F39 gpu watch · F40 gpu proof · F41 test list · F42 test
 matrix ·
 F43 benchmark compare · F44 endurance status · F45 endurance run · F46 report · F47 release gate ·
 F48 sarif export · F49 shell completion · F50 cockpit.

5. Command surface (CLI)

help, version, capabilities, status, project list|use|inspect|watch, drift, timeline,
 snapshot create|list|show, diff, goal proof, todo board, artifact list|show|search|verify,
 addon verify, config, doctor, diagnostics, preflight, health [--watch], models,
 model show|smoke|benchmark, route <class>, gpu|watch|proof, test list|matrix,
 benchmark compare <a> , endurance status|run <rung>, report, release gate, export sarif,
 completion powershell|bash|zsh, cockpit.

Global flags: --format=json|ndjson|tsv, --color=auto|always|never, --explain/--dry-run.

6. Packages & entry points

- cli-cpp/src/: pos_json.hpp (JSON parser), pos_model.hpp (registry/goal/todo), pos_runner.hpp (dispatch + parseCmdResult pure), pos_process.hpp (ProcessRunner), pos_terminal.hpp, pos_output.hpp, pos_exitcodes.hpp, pos_protocol.hpp, pos_health.hpp, project_os_cli.cpp (main/menu).
- cli-cpp/tests/test_pos.cpp (unit + fuzz), cli-cpp/CMakeLists.txt, cli-cpp/CMakePresets.json.
- bin/project-os-bridge.mjs (protocol v2: protocol/requestId/ok/status/result/timingMs/errors).

7. Honest limitations

- /backend/monitor returns 500 (signature mismatch) — GPU proof uses VRAM delta + HTTP 200.
- Endurance rung 30/60 pending external GPU VRAM (~3 GB) — reported honestly, never fake PASS.
- Model backend/quant/license UNKNOWN when not declared by LocalAI.
- Cockpit = inline VT renderer (no FTXUI; minimal dependency).
- No cloud, no telemetry, no source upload — loopback only.

8. Evidence pack (artifacts/cli-v3)

- release/CLI_V3_RELEASE_GATE.json (productionReady: true), release/CLI_V3_FEATURE_MATRIX.json
- CLI_V3_RELEASE_REPORT.md, CLI_V3_FEATURE_MATRIX.json (50/50),
CLI_V3_IMPLEMENTATION_PLAN.md,
CLI_V3_THREAT_MODEL.md, CLI_V3_PACKAGING.md, CLI_V3_SECURITY_REPORT.md,
CLI_V3_SOAK_REPORT.md,
research/CLI_V3_RESEARCH_CARDS.md.

9. Candidate focus areas for the next 50 phases (to be refined by ChatGPT web)

- Fuzz/property harness (libFuzzer), SARIF deeper, TUI upgrade (FTXUI decision gate), completion for
subcommands + dynamic slugs, protocol v3 negotiation, request correlation/trace, performance budget,
memory safety (sanitizers), packaging (CPack/installer, DEB/ZIP), CI matrix (Windows/Linux/macOS),
streaming/LocalAI parity, multi-model routing, endurance orchestration integration, diagnostics
redaction golden tests, Unicode Windows console (UTF-8 mode), config precedence source view,
status bar /
VS Code command parity, snapshot diff richer, workspace drift baseline compare, artifact store
integration,
budget/token integration, observability (OpenTelemetry local), accessibility (monochrome+small
terminal),
machine-consumer contract (schema version), exit-code taxonomy extension, and 50 real-user
scenarios.